

8.2 Confidence Interval on the Mean of a Normal Distribution (Variance Unknown)

In case of a **given** σ , the confidence interval for μ is determined using a random variable Z :

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \sim N(0; 1)$$

Since σ is **unknown**, σ is estimated using the **sample standard deviation** s . In this way, a random variable T arises:

$$T = \frac{\bar{X} - \mu}{S / \sqrt{n}}$$

This random variable T follows a **t distribution with $n - 1$ degrees of freedom (= df)**.

Remarks

1. Notation: $T \sim t(df)$
2. The **t distribution** is similar to the **standard normal distribution**, but is slightly flatter. The smaller the number of **degrees of freedom**, the flatter is the distribution.

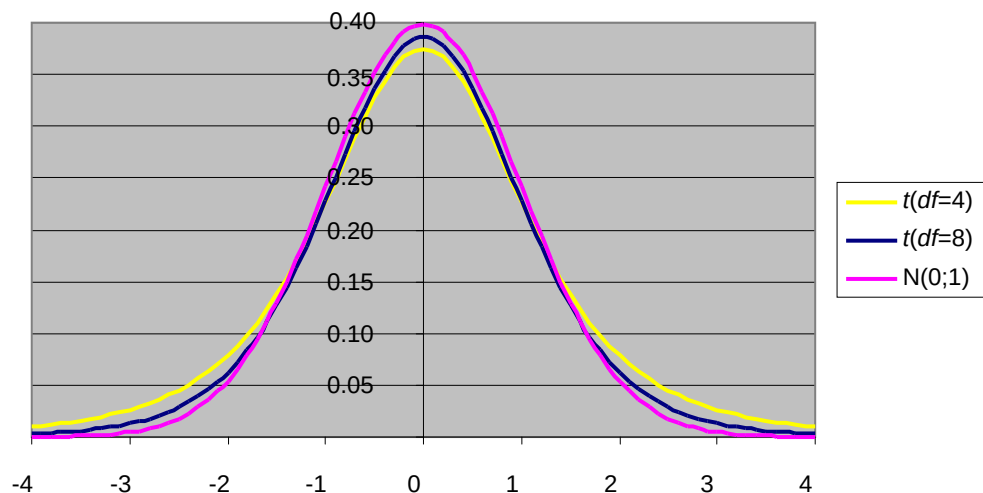


Figure 8.3

Probability density function of several t distributions

3. The following applies to the expected value and variance of a random variable $T \sim t(df)$:

- $E(T) = 0$
- $V(T) = \frac{df}{df-2}$

- For large values of df (i.e. $df > 40$), T has approximately the same properties as a **normal distribution**.

Therefore

$$T \approx N(\mu = 0; \sigma = \sqrt{\frac{df}{df-2}})$$

- Table V provides percentage points of the t distribution. It contains values of the random variable T above which we find a probability α .
It is also possible to use the **graphical calculator**. In that case, we choose the options **tcdf(a, b, df)** (i.e. **gc/distr/6**) and **invT(p, df)** (i.e. **gc/distr/4**).

Examples

- Let $T \sim t(df = 8)$. Then: $P(T \geq 2.306) = 0.025$ **Table V**
- Let $T \sim t(df = 8)$. Then: $\text{tcdf}(2.306, 10^{10}, 8) = 0.025$ **gc/distr/6**
- Let $T \sim t(df = 120)$. Then: $P(T \leq 1.980) = 0.975$ **Table V**
- Let $T \sim N(\mu = 0; \sigma = \sqrt{\frac{120}{118}} = 1.008)$.

Then: $P(T \leq 1.98) = \text{normalcdf}(-10^{10}, 1.98, 0, 1.008) = 0.9750$

Remarks

- The number 2.306 is denoted by **$t_{0.025; 8}$**
- On the basis of **symmetry**, **$t_{0.975; 8} = -2.306$**

Theorem

Suppose that $X_1, X_2, \dots, X_n$ is a random sample from a normal distribution with unknown mean μ and unknown standard deviation σ . Then a **$100(1 - \alpha)\%$ CI** for μ is given by:

$$[\vartheta_1; \vartheta_2] = \left[\bar{x} - t_{\frac{1}{2}\alpha; n-1} \cdot \frac{s}{\sqrt{n}}; \bar{x} + t_{\frac{1}{2}\alpha; n-1} \cdot \frac{s}{\sqrt{n}} \right]$$

where

$$F(t_{\frac{1}{2}\alpha; n-1}) = 1 - \frac{1}{2}\alpha, \quad \text{i.e.} \quad t_{\frac{1}{2}\alpha; n-1} = \text{invT}(1 - \frac{1}{2}\alpha, n - 1)$$

Proof

The assertion immediately follows from:

$$P(-t_{\frac{1}{2}\alpha; n-1} \leq T = \frac{\bar{X} - \mu}{s/\sqrt{n}} \leq t_{\frac{1}{2}\alpha; n-1}) = 1 - \alpha$$

Example 7

(see the data in Example 8-6)

Determine a 95%-CI for μ using the following measurements:

19.8	10.1	14.9	7.5	15.4	15.4	15.4	18.5	7.9	12.7	11.9
11.4	11.4	14.1	17.6	16.7	15.8	19.5	8.8	13.6	11.9	11.4

Solution

Using a graphical calculator:

$$\bar{x} = 13.7136$$

$$s = 3.5536$$

Since $n = 22$ and $\alpha = 0.05$:

$$t_{\frac{1}{2}\alpha; n-1} = t_{0.025; 21} = \text{invT}(.975, 21) = 2.080$$

[gc/distr/4](#)

So the resulting CI is

$$[\vartheta_1; \vartheta_2] = [13.7136 - 2.080 \cdot \frac{3.5536}{\sqrt{22}}; 13.7136 + 2.080 \cdot \frac{3.5536}{\sqrt{22}}] = [12.1378; 15.2895]$$

Remarks

1. Using the option [TInterval](#) of the graphical calculator:

$$[\vartheta_1; \vartheta_2] = [12.138; 15.289]$$

[gc/stat/tests/8](#)

2. A **CI** for μ with **given** $\sigma = 3.5536$ is always **shorter**:

$$[\vartheta_1; \vartheta_2] = [12.229; 15.199]$$

[gc/stat/tests/7](#)

More information therefore leads to a **more accurate** estimate.

3. Answering the question “**how large should n be to achieve a certain accuracy?**” is difficult, since $t_{\frac{1}{2}\alpha; n-1}$ depends on n .

In that case the procedure is as follows:

- Suppose $\sigma =$ the value determined for s .
- Determine the minimum sample size n , based upon a **ZInterval**. Say $n = n_0$.
- Using “**trial & error**” and using the option **TInterval**, determine n , where $n = n_0 + 1, n = n_0 + 2, \dots$.

Example 8

(see the data in Example 7)

How large must n be if the length of the 95% CI is to be 2?

Solution

Suppose $\sigma = 3.5536$. in that case $n_0 \geq \left(\frac{z_{\frac{1}{2}\alpha} \cdot \sigma}{E} \right)^2 = \left(\frac{1.9600 \cdot 3.5536}{\frac{1}{2} \cdot 2} \right)^2 = 48.5$. So $n_0 = 49$.

Using “trial & error” and the option TInterval of the graphical calculator:

- $n = 50$: $[\vartheta_1; \vartheta_2] = [12.704; 14.724]$
- $n = 51$: $[\vartheta_1; \vartheta_2] = [12.714; 14.713]$

So n must be 51.

Final remark

When determining a confidence interval for μ , it is assumed that the measurements on the basis of which the interval is calculated follow a **normal distribution**.

If the number of measurements is large, a **histogram** gives an answer to the question whether there is a normal distribution. If desired software packages such as **IBM SPSS Statistics** can draw a **normal curve** through this histogram.

If the number of measurements is small, we use a **normal probability plot** (a so-called **NP-plot**). See Section 6.7 in the textbook.

For a **normal probability plot** it is necessary that the measurements $x_1, x_2, \dots, x_n$ are arranged in ascending order: $x_{(1)}, x_{(2)}, \dots, x_{(n)}$ and that their cumulative frequencies $(i - 0.5)/n$ are calculated. See the ten measurements in **Example 6-7** in the textbook. It concerns a study into the operating time of laptop batteries. See also the data in Table 8.1.

Two **z-scores** play a role in a **normal probability plot**:

- the z-score of the measurement itself: $z_i = \frac{x_i - \bar{x}}{s}$
- the z-score of the cumulative frequency: $z_i = \text{invNorm}\left(\frac{i - 0.5}{n}, 0, 1\right)$

Check that the sample mean is given by 195.7 and that the sample standard deviation s equals 14.0321.

See the calculations shown in Table 8.1.

Table 8.1

Calculations for constructing a normal probability plot

i	$x_{(i)}$	z_i	$(i - 0.5)/10$	z_i
1	176	-1.4039	0.05	-1.6449
2	183	-0.9051	0.15	-1.0364
3	185	-0.7625	0.25	-0.6745
4	190	-0.4062	0.35	-0.3853
5	191	-0.3349	0.45	-0.1257
6	192	-0.2637	0.55	0.1257
7	201	0.3777	0.65	0.3853
8	205	0.6628	0.75	0.6745
9	214	1.3042	0.85	1.0364
10	220	1.7317	0.95	1.6449

A normal probability plot can be constructed¹ on ordinary axes by plotting both standardized scores z_i against $x_{(i)}$. This results in a straight line (i.e. the first z-score plotted against $x_{(i)}$) and a scatter plot (i.e. the second z-score plotted against $x_{(i)}$). If there is no clear difference between the scatter plot and the straight line, a normal distribution adequately describes the data. Rule

¹ Using Excel the normal probability plot in Figure 8.4 can be constructed as follows:

- Using Excel create a table whose contents are the same of those in Table 8.1.
- Cell A1 contains the text “i”.
- Use the formula “=(B2-AVERAGE(B\$2:B\$11))/STDEV(B\$2:B\$11)” to determine the outcome in cell C2.
- Copy the contents of cell C2 to cells C3 to C11.
- Use the formula “=(A2-0.5)/10” to determine the outcome in cell D2.
- Copy the contents of cell D2 to cells D3 to D11.
- Use the formula “=NORMINV(D2;0;1)” to determine the outcome in cell E2.
- Copy the contents of cell E2 to cells E3 to E11.
- Choose the *Insert* tab in the main menu.
- Choose *Scatter*, followed by *Scatter with only Markers*.
- Drag the open window to an empty space next to the table with the data to be used (i.e. Table 8.1; cell A1 contains “i”).
- Click with the right mouse button on the empty window and choose *Select Data*.
- Choose *Add*. In *Series X values* select cells B2:B11, in *Series Y values* (after removing “={1}”) cells C2:C11. Finish with *OK*.
- Choose *Add* again. Select cells B2:B11 and E2:E11 respectively.
- Delete the *legend entries* “Series 1” and “Series 2”.
- Click with the right mouse button on the horizontal axis and choose *Format Axis*. Choose *Axis Options*. *Minimum*, *Maximum* and *Major unit* must be *Fixed*. Fill in the values 170, 230 and 10 respectively.
- Click with the right mouse button on the vertical axis and choose *Format Axis*. Choose *Number* followed by *Number* and enter the number 1 at *Decimal places*.
- Select the points that form a straight line. Select *Add Trendline* with the right mouse button. The *Trendline Options* are *Linear*, *Forward* = 3 and *Backward* = 3. At *Line Color* choose *Solid line* with a blue *Color*. At *Line Style* choose *Width 2*.
- Click with the right mouse button on a blue dot. Choose *Format Data Series*, followed by *Marker Options*. Tick the option *None*.
- If necessary, remove the horizontal grid lines (select the lines, then press delete).

of thumb: imagine a “fat pencil” lying along the straight line. If all points are covered by this pencil, a normal distribution is likely.

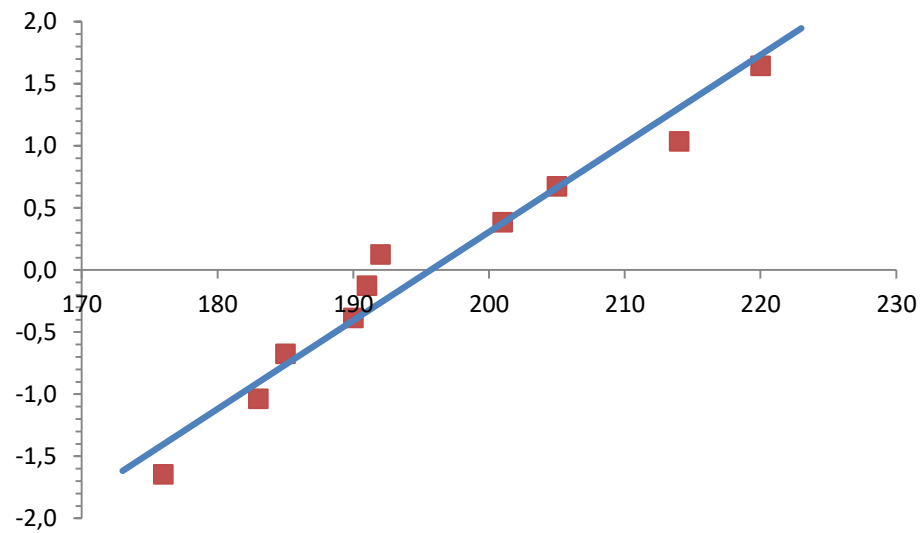


Figure 8.4 Normal probability plot obtained from standardized normal scores

Thus, in the case of the operating time of laptop batteries, there is more or less a normal distribution.